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In [1]: from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
    channel="ibm_quantum_platform",
    token="dW13vI9wd6XvAyxsxbb0Knc-lKnUSbXKg8VZLS8QHVMk",
    overwrite=True
)
```

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In [2]: service = QiskitRuntimeService()
backends = service.backends()
print([b.name for b in backends])
```

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IBMInputValueError                                Traceback (most recent call last)
Cell In[2], line 1
----> 1 service = QiskitRuntimeService()
      2 backends = service.backends()
      3 print([b.name for b in backends])

File ~/anaconda3/envs/qml/lib/python3.11/site-packages/qiskit_ibm_runtime/qi
skit_runtime_service.py:254, in QiskitRuntimeService.__init__(self, channel,
token, url, filename, name, instance, proxies, verify, private_endpoint, url
_resolver, region, plans_preference, tags)
    252 else:
    253     self._api_clients = {}
--> 254     instance_backends = self._resolve_cloud_instances(instance)
    255     instance_names = [instance.get("name") for instance in self._bac
kend_instance_groups]
    256     instance_plan_names = {
    257         instance.get("plan") for instance in self._backend_instance_
groups
    258     }

File ~/anaconda3/envs/qml/lib/python3.11/site-packages/qiskit_ibm_runtime/qi
skit_runtime_service.py:685, in QiskitRuntimeService._resolve_cloud_instanc
es(self, instance)
    673 if not self._backend_instance_groups:
    674     self._backend_instance_groups = [
    675         {
    676             "name": inst["name"],
    (...) 683             for inst in self._all_instances
    684         }
--> 685     self._filter_instances_by_saved_preferences()
    687 return [(inst["crn"], inst["backends"]) for inst in self._backend_in
stance_groups]

File ~/anaconda3/envs/qml/lib/python3.11/site-packages/qiskit_ibm_runtime/qi
skit_runtime_service.py:364, in QiskitRuntimeService._filter_instances_by_sa
ved_preferences(self)
    362 if self._plans_preference:
    363     error_string += f"plan: {self._plans_preference}"
--> 364 raise IBMInputValueError(
    365     "No matching instances found for the following filters:",
    366     f"{error_string}.",
    367 )

IBMInputValueError: 'No matching instances found for the following filters:
.'
```

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In [10]: from qiskit import QuantumCircuit, transpile
        from qiskit_ibm_runtime import SamplerV2 as Sampler, QiskitRuntimeService

        # Circuit bauen
        qc = QuantumCircuit(2, 2)
        qc.h(0)
        qc.cx(0, 1)
        qc.measure([0, 1], [0, 1])
```

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# Verbinden
service = QiskitRuntimeService()
backend = service.backend("ibm_kingston")

# Transpilieren (auf Backend-native Gates übersetzen)
qc_transpiled = transpile(qc, backend=backend)

# Job abschicken
sampler = Sampler(backend)
job = sampler.run([qc_transpiled], shots=1024)

print("Job ID:", job.job_id())
print("Status:", job.status())

```

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qiskit_runtime_service.__init__:WARNING:2026-04-27 22:21:45,010: Instance was not set at service instantiation. Free and trial plan instances will be prioritized. Based on the following filters: (tags: None, region: us-east, eu-de), and available plans: (open), the available account instances are: open-instance. If you need a specific instance set it explicitly either by using a saved account with a saved default instance or passing it in directly to QiskitRuntimeService().
qiskit_runtime_service.backends:WARNING:2026-04-27 22:21:45,012: Using instance: open-instance, plan: open
Job ID: d7nsan497osc73dsfbr0
Status: RUNNING

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In [11]: # Ergebnis abrufen (blockiert bis Job fertig ist)
result = job.result()

# Counts auslesen
counts = result[0].data.c.get_counts()
print("Ergebnisse:", counts)

```

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Ergebnisse: {'00': 499, '11': 514, '10': 7, '01': 4}

```